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High-resolution assessment of wind energy potential in the Hami region of Northwestern China

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E-mail: zengzz@sustech.edu.cn**Keywords:** renewable energy, wind power potential, advanced wind turbine technology, interannual variabilitySupplementary material for this article is available [online](#)

Abstract

Wind energy plays a pivotal role in the global effort to mitigate climate change, with China emerging as a leader in renewable energy adoption. The Hami region in northwestern China stands out as a crucial area for wind power development, given its substantial wind resources and strategic importance in China's energy landscape. However, existing studies on wind energy potential vary widely and involve large uncertainties due to sparse measurements and coarse resolution, highlighting the need for more precise assessments to guide policy decisions and optimize energy utilization. This study leverages high-resolution ERA5 reanalysis data and advanced wind turbine technology to assess wind energy potential in the Hami region, taking into account factors including wind speed patterns, turbine heights, and geographical constraints. The comparison with *in-situ* data demonstrated that high-resolution ERA5 reanalyzed wind speeds enable to capture multi-year wind speed variations in this region. We find substantial potential for wind energy in Hami, with energy densities exceeding 200 W m^{-2} in the high-potential wind zones. Importantly, this study identifies a new high-potential area in eastern Hami, termed the East Wind Zone. Our high-resolution assessment of wind energy potential at different heights over the past two decades reveals long-term trends and seasonal variations. Increasing the hub height from 95 m to 140 m raises the average wind power generation potential across Hami by 31.3 GWh yr^{-1} . Our findings highlight the importance of strategic wind farm placement to maximize renewable energy output and provide insights for policy and industry, supporting China's renewable energy goals.

1. Introduction

Wind energy, characterized by low greenhouse gas emissions and minimal air pollution, has become a prominent component of the renewable energy system due to its mature industrial technology and cost-effectiveness (Ling and Cai 2012, Islam *et al* 2013, Belabes *et al* 2015). In alignment with the goals outlined in the 2015 Paris Agreement to mitigate

global warming (UNFCCC 2015), China has increasingly transitioned towards renewable energies, with a particular emphasis on the abovementioned wind energy. Since 2009, China has maintained the position of the world's largest installed wind power capacity (Zhang *et al* 2020), and reached a cumulative capacity of 441 GW by the end of 2023 (GWEC 2024). In addition, aiming to achieve a target wind capacity of 600 GW by 2030 (Wang *et al* 2022), a considerable

number of wind farms have been planned and are currently under construction.

However, the regional utilization of wind energy in China exhibits highly heterogeneous and its assessment still involves significant uncertainties. China's vast territory and diverse topography result in uneven distribution of wind resources across different regions, presenting challenges for accurately modeling and assessing wind energy potential. According to China's 14th 'Five-Year Plan' (2021–2025), Northwestern China is recognized for its great wind energy development potential (State Council of the People's Republic of China 2021). The Hami region, east of the Tianshan Mountains, is designated as the most important area for wind power generation. Hami possesses a land resource suitable for the construction of large-scale wind power plants because of its desert and Gobi landscape. By the end of 2023, Hami's installed grid-connected wind power capacity reached 13.1 GW (Wusierhan 2023). Furthermore, wind power in Hami also significantly contributes to the transmission of excessive electricity to other regions of China (Tao 2022). For example, the electricity transmitted to Henan Province in central China reached 47.2 TWh in 2023, reducing Henan's CO₂ emissions by 42 million tons (Li 2024). Similarly, the Hami-Chongqing ±800 kV ultra-high-voltage direct current transmission project is projected to deliver over 36.0 TWh of electricity per year to Chongqing in southwestern China. Therefore, given the critical role of the Hami region in China's renewable energy landscape, obtaining precise wind energy assessments in this region is imperative for optimizing resource allocation, informing policy decisions, guiding strategic planning, and promoting sustainable development.

Wind energy potential determines the availability and feasibility of wind energy utilization, and hence is important for the site selection and return on investment of new wind farms. Since around 2010, there has been a significant upward trend in surface wind speeds (Zeng *et al* 2019, Liu *et al* 2022), causing a substantial impact on wind energy generation as it is directly proportional to the cube of wind speed (Lu *et al* 2009). Moreover, with the increasing installed capacity of wind turbines, the total wind energy generation has become even more sensitive to variations in wind speed. Therefore, accurately assessing wind energy potential is crucial. A high-resolution assessment of wind energy potential facilitates a more precise comprehension of the spatiotemporal characteristics of wind resources and could provide the scientific basis for the government to formulate policies and optimize the energy mix.

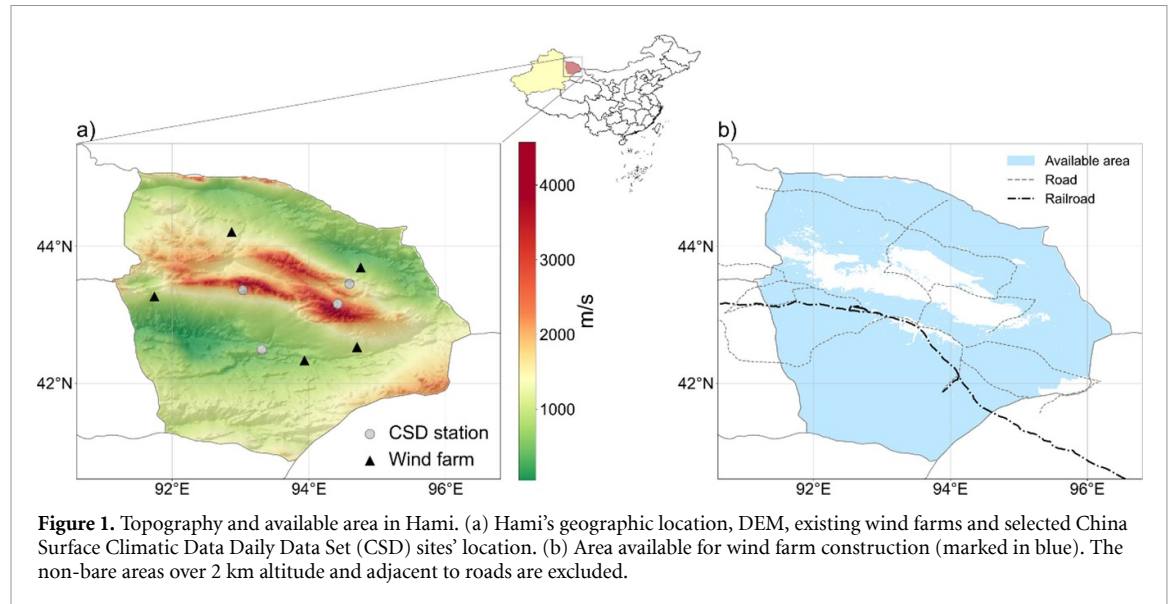
A large number of studies have assessed wind energy potential on both country-level and regional scales, but their results vary considerably. McElroy *et al* (2009) used GEOS-5 data to assess China's wind potential as 24.7 PWh, while Davidson *et al* (2016) used the same data to obtain a result of only 2.6 PWh. Dupont *et al* (2018) used the ERA5-Interim dataset with a grid size of $0.75^\circ \times 0.75^\circ$ to assess wind energy potential at different hub heights (71 m and 124 m) and the result for China was 6.2 PWh yr^{-1} . Li *et al* (2016) studied the wind energy resources at 100 m hub height in the Bohai and Yellow Seas of China using the ERA-Interim reanalysis dataset. They found that the wind power density in the coastal areas is 300 W m^{-2} to 500 W m^{-2} . Liu *et al* (2019) considered that the observation wind speed had better accuracy and chose 2430 climate stations to divide the whole of China into a grid of $1.5^\circ \times 1.5^\circ$ to construct a wind energy density distribution map and found the national average annual wind power density to be 164.1 W m^{-2} .

The large variations in the results of the previous studies are driven by a combination of factors, including differences in data resolution, which affect the ability to capture the impact of topographic and geomorphic features on wind speed distribution. Low-resolution grids may fail to capture the changes caused by these micro-geographic differences. In addition, assumptions for wind turbine technology have a great impact on wind energy output, especially the designed hub height and setting density. Moreover, the definition of geographical restrictions is an important factor for available areas. There is a large body of research that binarily divides grid cells into restricted and unrestricted areas (He *et al* 2014, Liu *et al* 2017, Jung *et al* 2022), as well as studies (Herran *et al* 2016, Lopez *et al* 2021) that explicitly quantify the impact of geographic constraints and barriers on the results of wind energy potential. However, these factors in the Hami region have not been adequately considered before. The unique topographic, geomorphic, and technological conditions in Hami necessitate a more detailed analysis to accurately evaluate its wind energy potential.

To address this research gap, we utilize the high-resolution European Centre for Medium-Range Weather Forecasts atmospheric reanalysis dataset to improve the accuracy of the wind potential assessment in the Hami region. Additionally, we incorporate the latest wind turbine technology and local land-use policies to provide a detailed assessment of wind energy potential at various hub heights. By integrating these factors, our study aims to deliver a more precise and localized evaluation of wind energy potential in the Hami region.

Table 1. Information on selected CSD stations.

ID	Location	Latitude (N)	Longitude (E)	Elevation (m)	Anemometer height (m)
1	Balikun	43.36	93.03	1679.4	1680.4
2	Naomaohu	43.45	94.59	479.0	480.0
3	Yiwu	43.16	94.42	1728.6	1729.6
4	Hami	42.49	93.31	737.2	738.3



2. Data and methodology

2.1. Wind speed data

The wind speed data used to calculate wind energy potential was adopted from the ERA5 reanalysis dataset (Hersbach *et al* 2020) (URL: <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land?tab=overview>), including transverse (u) and meridional (v) wind speed at a height of 10 m (U_{10}) and 100 m (U_{100}) above ground from 1981 to 2022. The vector wind speed data has a spatial resolution of $0.1^\circ \times 0.1^\circ$ and an hourly temporal resolution. In addition, hourly *in-situ* wind observations provided by the China Surface Climatic Data Daily Data Set (CSD) (URL: <http://data.cma.cn/en/?r=data/>) were used to evaluate the reliability of the reanalysis data. Based on the quality control recommended by the World Meteorological Organization, four CSD stations in the study area were selected (shown in table 1 and figure 1(a)) that met the two criteria: (1) <11 missing daily values in a month and (2) <5 consecutive missing daily values in a month. Seasonal statistics cover the following months: spring (March–May), summer (June–August), fall (September–November), and winter (December–February).

The wind speed at the hub height (H) of the turbine is derived from the power law wind profile (Jung *et al* 2021, Yang *et al* 2024). The power law exponent

α is calculated by U_{10} and U_{100} for each grid:

$$\alpha = \frac{\ln\left(\frac{U_{100}}{U_{10}}\right)}{\ln\left(\frac{100\text{m}}{10\text{m}}\right)}. \quad (1)$$

Then the wind speed could be extracted to hub height H from U_{10} :

$$U_H = U_{10} \times \left(\frac{H}{10\text{m}}\right)^\alpha. \quad (2)$$

2.2. Wind speed distribution

To minimize uncertainty in wind resource assessment, wind power density and turbine power should be derived from appropriate wind speed probability distributions. As the most commonly used distribution, the Weibull distribution has been proven to be appropriate for modeling wind speed in China (Wu *et al* 2013, Li *et al* 2018). Here we used the Weibull function to simulate the distribution of wind speed. Specifically, the Weibull distribution consists of two parameters: a shape parameter k and a scale parameter λ (Manwell *et al* 2010), with its distribution function listed below:

$$f(U; \lambda, k) = \frac{kU^{k-1}}{\lambda^k} \exp\left[-\left(\frac{U}{\lambda}\right)^k\right] \quad (3)$$

$$F(U; \lambda, k) = 1 - \exp\left[-\left(\frac{U}{\lambda}\right)^k\right], \quad (4)$$

where f and F denote the probability density function and Cumulative distribution function, respectively.

Table 2. Geographical restrictions.

Restriction area	Distance/Threshold	Horizontal resolution
Non-bare land		30 m
High elevation	>2000 m	100 m
Slope	>15°	100 m
Road/railroad	300 m buffer	10 m

2.3. Geographical restrictions

Evaluating wind power potential requires identifying the extent of land available for wind power development. The land surface data collected in this study consisted of four sources. The land use data was selected from the 2022 land-use distribution in the ESA CCI (URL: <https://cds.climate.copernicus.eu/datasets/satellite-land-cover?tab=download>) with a spatial resolution of 300 m. It was further interpolated to the same resolution of $0.1^\circ \times 0.1^\circ$ as the surface wind speed data. The road network data was provided by OpenStreetMap. The Digital Elevation Model (DEM) was provided by Cold and Arid Regions Science Data Center, Lanzhou, China (URL: www.re3data.org/repository/r3d100010297), and protected area data was provided by The World Database on Protected Areas, (WDPA, URL: www.protectedplanet.net/en/thematic-areas/wdpa?tab=WDPA).

Previous studies have adopted two primary methods for screening available land areas. The first method uses a binary classification, dividing grid cells into restricted and unrestricted areas (He and Kammen 2014, Liu et al 2017, Jung and Schindler 2022). The second method assigns coefficients to different land use types to reflect the varying degrees of suitability for wind energy development (Herran et al 2016, Lopez et al 2021). In this study, we refer to the requirements of ‘*Methodology of wind energy resource assessment for wind farm* (GB/T 18710-2002)’, selecting bare land as the available area and excluding areas within 300 m of roads or cities. Due to the vast and sparsely populated nature of the Hami region, where bare land constitutes a large proportion, this method of selecting available areas minimizes environmental impact. In addition, based on the existing construction conditions of the wind turbine, the restricted areas also include areas with an altitude of more than 2000 m and a slope of more than 15° (table 2). The delineation of available regions for turbine construction, after the exclusion of geographical constraints, is presented in figure 1(b).

2.4. Assessment of wind energy potential

Three typical wind turbines, with a rated capacity of 5 MW at different hub heights (95 m/120 m/140 m), were chosen to estimate the wind energy potential in the Hami region. The selected turbines represent the highest level of current technology in China and the

future development trend, providing a more accurate estimate of wind energy potential. The turbines are manufactured by Gamesa and Multibrid, which provide power curves $P(U)$ for the corresponding turbines (figure 2). Based on the power curve and wind speed distribution, the wind energy output for a year (E) could be calculated as the following integral equation:

$$E = T \times \int_0^{\infty} P(U) \cdot f(U) dU, \quad (5)$$

where T is the number of hours in a year.

Many existing studies for modeling wind farm layouts are oversimplified, lacking detailed simulations reflecting real installation scenarios (Bosch et al 2017, Eureka et al 2017, Dupont et al 2018). This study considers the actual layout of the turbines to achieve a more accurate assessment of the wind energy potential. According to the guidelines of the ‘*Code for design of wind farm* (SAC 2015)’, each wind turbine covers an area of 1 km², with a spacing of 2 km between wind farms. This method not only adheres to minimize the wake effect of the turbine but also ensures a more precise representation of real-world conditions in the Hami region.

3. Results

3.1. Wind speed validation

To assess the accuracy of the reanalysis dataset in Hami, we validated the dataset against observational wind speed data from the CSD site. Notably, the wind speed recorded from the CSD sites is available up to the year 2018, whereas ERA5 offers a comprehensive time series spanning from 1980 to 2022. The monthly and annual variations of wind speeds in Hami are depicted in figure 3(a). Examining the inter-annual variations, it is evident that the reanalyzed data has no significant fluctuations over the entire temporal range. It can be attributed to the inherent smoothing effects of the numerical models used in reanalysis and the data assimilation process (Torrallba et al 2017, Ramon et al 2019, Wohland et al 2019). In addition, ERA5 fails to capture the declining trend observed in station data before the year 2000. Further scrutiny reveals that the reanalysis of wind speed for 1992 and 1993 displays unusually low values, which are considered to be systematic errors and are excluded from the dataset.

The analysis of wind speed distribution from CSD station data and ERA5 reanalysis data before and after the year 2000 reveals that between 1981 and 2000, the annual and monthly reanalysis wind speeds were consistently lower than the observed wind speeds (figure 3(b)). However, from 2000 to 2018, the wind speeds from both datasets were relatively consistent (figure 3(c)). To further validate the significance of these differences, a T -test was conducted on the

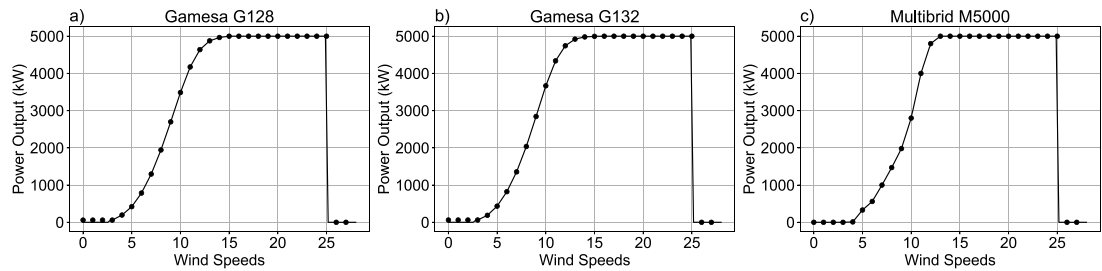


Figure 2. Properties of wind turbine power curves. The power curve of (a) Gamesa G128-5.0 MW wind turbine. (b) Gamesa G132-5.0 MW wind turbine. (c) Multibrid M5000 wind turbine.

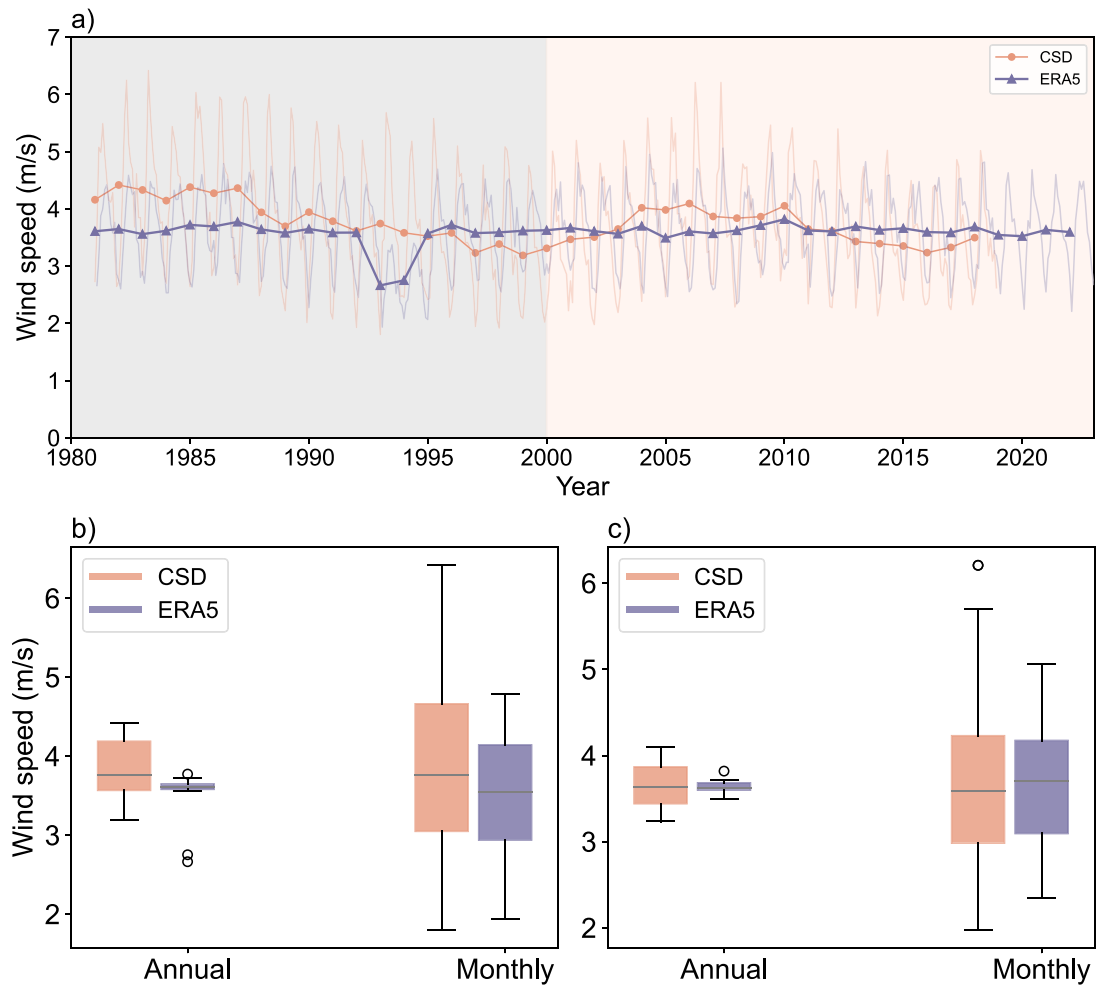


Figure 3. Validation of wind speed in the Hami region. (a) Time series of wind speed from CSD stations and ERA5 reanalysis dataset over the period 1981–2022. (b) Boxplot compares wind speeds at CSD stations and the ERA5 reanalysis dataset between 1981 and 2000. (c) Boxplot comparing wind speeds at CSD stations and ERA5 reanalysis dataset between 2000 and 2022.

monthly and annual average wind speeds from the CSD and ERA5 datasets before and after 2000. The results indicate a significant difference ($P < 0.05$; table 3) between the ERA5 and CSD station wind speeds before 2000, while no significant deviation ($P > 0.05$; table 3) was found between the ERA5 and observed wind speeds after 2000. Consequently, we selected the ERA5 reanalysis dataset from the post-2000 period for subsequent assessments in this study.

3.2. Wind speed at hub height

The wind speeds at 10 m in Hami reveal that wind speeds are higher in the northern and western parts of the region and lower in the central part, where the Tianshan Mountain range is located (figure 4(a)). In calculating wind speeds at different heights (equation (1)), the power-law exponent (α) at most of the grids varies considerably from an empirical value of $1/7$ (figure 4(b)), which is

Table 3. Evaluated p -value of T -test between observational and reanalysis wind speed. p -value > 0.05 indicates no significant difference.

p -Value	1981–2000	2000–2018
Monthly	0.001 (<0.05)	0.80 (>0.05)
Annual	0.01 (<0.05)	0.75 (>0.05)

commonly utilized in the field of engineering (Tizpar *et al* 2014, Belabes *et al* 2015, Boopathi *et al* 2021). This is because the observed power-law exponent α can vary spatiotemporally due to atmospheric stability and surface roughness conditions (Gualtieri *et al* 2019, Yang *et al* 2024). Therefore, it is necessary to use specific values for different grids. Based on our results, the region with the highest power-law exponent ($\alpha > 0.25$) is located in the Tianshan Mountains (figure 4(b)), indicating the complexity of the terrain at high altitudes, where steep mountain ranges lead to dramatic vertical changes in atmospheric flow and vertical direction, and thus causing the high power-law exponent of the wind profile. By contrast, the plains in the south of the Tianshan Mountains show a negative power-law exponent, down to about -0.2 . The anomalies resulted from the possible existence of special topographic features such as basins or valleys. The narrow topography causes the surface wind speed flow velocity to become larger and higher than the wind speed at high altitudes, resulting in a negative power law index.

Based on the wind speed at 10 m and the corresponding power-law exponent, we obtained the wind speed at hub heights (equation (2)) of 95 m, 120 m, and 140 m in Hami, respectively (figures 4(c)–(e)). The analysis of these figures reveals a discernible increase in wind speed with vertical height, with average wind speeds of 4.91 m s^{-1} , 5.02 m s^{-1} , and 5.18 m s^{-1} at the respective hub heights of 95 m, 120 m, and 140 m. The wind speed in different regions of Hami exhibits significant disparities. Specifically, within the central region, the wind speeds across different heights consistently fall below the typical cut-in speed for wind turbines (2.5 m s^{-1}), suggesting sub-optimal conditions for energy extraction. In comparison, the northern, western, and southeastern regions exhibit relatively higher wind speeds. Notably, not all of the locations of the existing wind farms in Hami have been chosen to be in the area of maximum wind speed, as indicated in figures 4(c)–(e). While it might be attributed to factors like construction costs and transmission convenience considered during the site selection process, there remains substantial potential for enhancement in the strategic placement of wind farms in the Hami region.

3.3. Wind energy density

According to the Chinese national standard for wind power generation, 10 m wind is pervasively utilized

to assess wind energy density (SAC 2002). The distribution of wind power density in the Hami region (figure 5(a)) indicates that the western and north-eastern regions consistently exhibit high wind power potential surpassing 200 W m^{-2} , among China's higher wind energy densities. In contrast, the south-eastern region and surrounding areas of the high-potential zones show a wind power potential ranging from 100 W m^{-2} to 200 W m^{-2} . Seasonal variations of wind power potential from 2000 to 2022 reveal a recurring pattern, with peak wind power potential occurring in April or May, reaching approximately 125 W m^{-2} , and troughs in December or January, plummeting to about 25 W m^{-2} (figure 5(b)).

These results suggest that the Hami region has an enormous wind resource with substantial potential for wind power generation. If wind farms are strategically located in high-potential zones, their output could be approximately tripled, emphasizing the importance of implementing refined site selection strategies to optimize deployment efficiency. Moreover, the findings demonstrate a cyclic variation in wind patterns. This cyclic behavior likely correlates with seasonal changes in atmospheric conditions influencing wind speeds. Considering these patterns, energy consumption practices can be adjusted during periods of high wind potential, such as prioritizing energy-intensive operations in April or May. The objective is to align energy consumption activities with periods of optimal wind power availability, thereby maximizing the utilization of renewable energy resources and improving overall energy system efficiency.

3.4. Wind energy potential

Wind power potential assessments for the Hami region were conducted across various hub heights after excluding geographical constraints, utilizing high-resolution wind speed reanalysis data (equation (5)). Figures 6(a)–(c) illustrate the long-term mean wind power potential at hub heights of 95 m, 120 m and 140 m, respectively. The analysis reveals that areas with high wind energy potential are predominantly located in the eastern part of Hami and the northern region of the Tianshan Mountains. These areas coincide with the three major planned wind zones in Hami (Zhang 2008): (1) Santanghu Wind Zone, (2) Baili Wind Zone, and (3) Southeast Wind Zone. Notably, this study identifies a previously unrecognized high-potential area in the eastern part of Hami. The wind energy potential in this region surpasses that of the Baili and Southeast Wind Zones on a per-unit-area basis. Consequently, this area has been defined as the fourth major wind zone in Hami, referred to as the East Wind Zone.

Increasing the hub height from 95 m to 140 m raises the average wind power generation potential across Hami by 31.3 GWh yr^{-1} (figure 6(d)). Specifically, the wind energy potential in the four

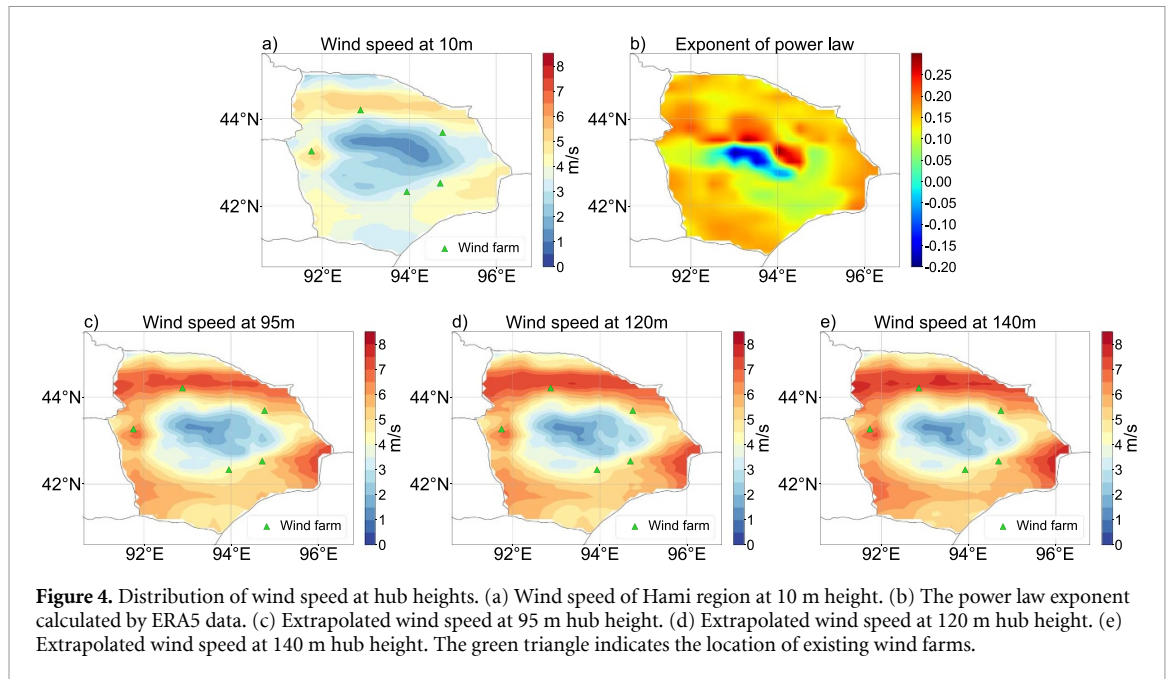


Figure 4. Distribution of wind speed at hub heights. (a) Wind speed of Hami region at 10 m height. (b) The power law exponent calculated by ERA5 data. (c) Extrapolated wind speed at 95 m hub height. (d) Extrapolated wind speed at 120 m hub height. (e) Extrapolated wind speed at 140 m hub height. The green triangle indicates the location of existing wind farms.

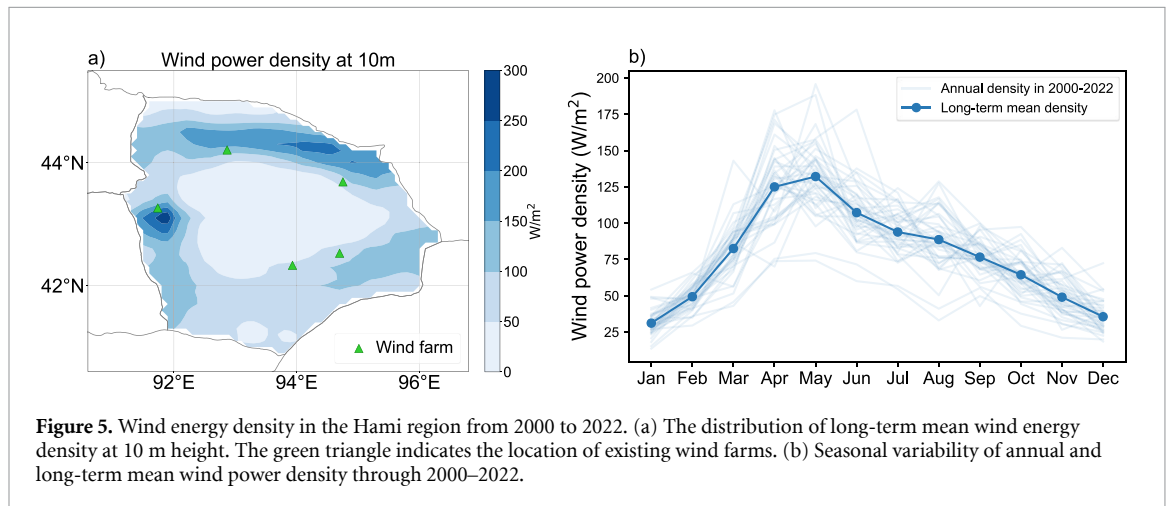


Figure 5. Wind energy density in the Hami region from 2000 to 2022. (a) The distribution of long-term mean wind energy density at 10 m height. The green triangle indicates the location of existing wind farms. (b) Seasonal variability of annual and long-term mean wind power density through 2000–2022.

major wind zones (figure 6(e)) increases by 19.4%, 14.0%, 22.1%, and 21.7%, respectively. Increasing the hub height yields substantial power generation benefits in areas with a higher power law exponent (e.g. East and Southeast Wind Zone), where surface friction is significant. Conversely, in the Baili Wind Zone, where the power law exponent is lower, and strong winds primarily result from the canyon topography between two mountains, the benefits of increasing the hub height are relatively lower. Therefore, when constructing wind turbines, the cost-effectiveness of different regions must be considered.

A clear trend indicates that as the hub height increases, the power generation capacity of wind turbines significantly improves (figure 6(f)). This suggests that technological advancements, particularly the increase in wind turbine hub height, have effectively enhanced wind power generation in the

analyzed region. While overall interannual variations remain modest during the period 2000–2022, there is a distinct seasonal pattern in generation potential. Seasonal trends reveal relatively consistent across the Hami region and (figure 6(g)), the four main wind zones (figure S1), with higher generations exhibiting in spring (MAM) and summer (JJA) and lower in autumn (SON) and winter (DJF). The peak season typically occurs in May with a multi-year average of 447.0 TWh at 95 m (473.3 TWh, 492.0 TWh at 120 m and 140 m, respectively). In contrast, the low season is observed in January, with a multi-year average of 134.6 TWh at 95 m (146.1 TWh, 154.1 TWh at 120 m and 140 m, respectively). Scientifically, these patterns align with expected wind characteristics related to seasonal atmospheric changes, influencing wind speeds and subsequently, electricity generation.

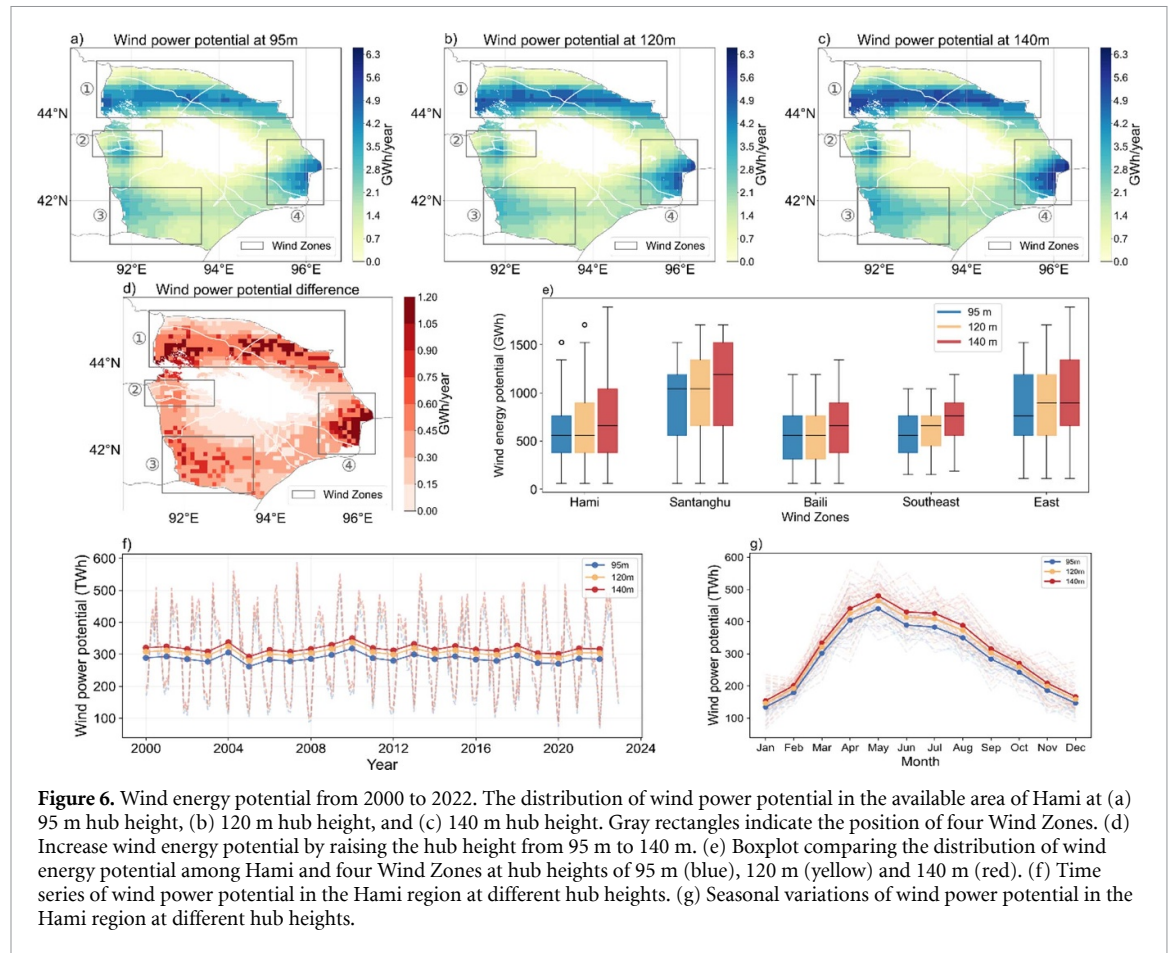


Figure 6. Wind energy potential from 2000 to 2022. The distribution of wind power potential in the available area of Hami at (a) 95 m hub height, (b) 120 m hub height, and (c) 140 m hub height. Gray rectangles indicate the position of four Wind Zones. (d) Increase wind energy potential by raising the hub height from 95 m to 140 m. (e) Boxplot comparing the distribution of wind energy potential among Hami and four Wind Zones at hub heights of 95 m (blue), 120 m (yellow) and 140 m (red). (f) Time series of wind power potential in the Hami region at different hub heights. (g) Seasonal variations of wind power potential in the Hami region at different hub heights.

4. Discussion

The assessment of wind energy potential involves several factors that contribute to uncertainties in the results. The most important factor is the resolution of wind speed data. This study utilized $0.1^\circ \times 0.1^\circ$ high-resolution wind speed data from ERA5 and validated it with observational data. It is essential to acknowledge the potential inaccuracies may arise when comparing reanalysis data with observed station data in wind energy potential assessments. Multiple reanalysis datasets, including ERA5, are unable to accurately capture trends in surface wind speed changes due to the insufficient simulation of complex terrain and atmospheric boundary layer conditions during the data assimilation process (Vautard *et al* 2010). In this study, we selected ERA5 wind speed product, which aligns most closely with climatological observations in Asia (Fan *et al* 2021). The systematic underestimation of wind speeds by ERA5 before 2000 may be attributed to limitations in the reanalysis model's representation of atmospheric dynamics, such as wind shear and turbulence, in earlier periods (Potisomporn *et al* 2023). These errors could potentially affect the reliability of long-term wind energy assessments, particularly if pre-2000 data were used.

To mitigate these issues, we opted to use ERA5 reanalysis data only from the post-2000 period, where wind speed has no significant difference from observed values, as demonstrated by the *T*-test results. Despite the reanalysis data may contain some errors over specific periods, it remains the most commonly used source for wind speed data in wind energy potential assessments due to its high spatial resolution. For example, He and Kammen (2014) and Liu *et al* (2019) suggested that Xinjiang has wind energy potential from 371.0 TWh to 691.0 TWh, based on the assessment of 200 sites and 2430 sites, respectively. In contrast, our study estimates wind energy potential in the Hami region alone to range from 286.8 TWh (95 m) to 318.1 TWh (140 m) per year (figure 6). Given that the Hami region occupies less than 10% of Xinjiang's area, this suggests a significantly higher per-area wind energy potential compared to previous studies of the entire Xinjiang region. Such underestimations can be attributed to the uneven spatial distribution of wind speed stations, especially in the sparsely covered northwest region of China with abundant wind resources. Furthermore, this discrepancy likely arises from the higher density of wind turbines used in our study, based on national standards, compared to the lower density employed in previous

studies. The increased turbine density directly contributes to the higher estimated potential.

The delineation of the four major wind zones, particularly the identification of the East Wind Zone, holds significant implications for wind energy development in the area. This newly identified high-potential region offers a substantial opportunity to enhance the overall wind energy output of Hami. By expanding the scope of development to include this area, it is possible to distribute wind turbines more effectively, thus optimizing the use of available wind resources. Wind energy potential in the four major wind zones increases with higher hub heights (figure 6(d)), particularly in zones with a higher power law exponent such as Santanghu and the East Wind Zones (figure 4(b)). Supplementary analysis (figure S2) reveals that increasing the hub height from 95 m to 120 m yields a notable increase in wind energy potential, though less than the increase observed when the hub height is further raised to 140 m. However, in areas with high wind shear values, the costs associated with increasing hub height are higher (Rehman *et al* 2013). The incremental cost of raising the hub height from 120 m to 140 m is typically greater than the initial increase from 95 m to 120 m. It is worth noting that a key limitation of this study is its focus on the technical potential of wind energy in Hami, without considering the economic impacts. In regions with high or negative wind shear, the costs of turbine construction and maintenance could be significantly higher, requiring more precise cost assessments. Future research should incorporate a comprehensive cost-benefit analysis to address this, especially by using more localized observational data to complement the reanalysis data used in this study. This would improve the accuracy of both wind energy potential and cost assessments, allowing for a more robust evaluation of the economic viability of wind energy development.

Given the substantial untapped potential for wind energy generation in the Hami region, the strategic siting of wind farms is a critical policy consideration. The suboptimal location of wind farms, as depicted in figures 4(c)–(e), suggests that factors such as construction costs and transmission convenience have taken precedence over the maximization of wind power generation during the site selection process. The policy implications of the study's findings underscore the need for a more nuanced approach to wind farm placement. The Hami region has three significant wind areas: the Santanghu wind area in northeastern Hami, the Baili wind area in western Hami, and the Southeast wind area. These regions have abundant year-round wind resources (figures 6(a)–(c)), and their stable wind speeds and high wind power densities provide ideal conditions for wind power generation. However, the challenging terrain and environmental conditions, such as the high altitude and

harsh climate of the Baili wind area, along with the issues of sandstorms and thunderstorms, necessitate careful consideration in site selection. Technological advancements have been crucial to address the challenges of environmental harshness and the variability of wind energy. Improved climate observation and forecasting and advancements in wind turbine technology have mitigated some environmental impacts (Douvi and Douvi 2023). Additionally, establishing energy storage facilities, such as the pumped-storage hydropower stations in Hami (Li *et al* 2023), has provided a means to balance the intermittency of wind power generation, thereby reducing energy losses and enhancing the stability of the power supply. The implementation of pumped-storage hydropower systems has demonstrated potential in smoothing the variability of wind energy, but its effectiveness depends on factors such as geographical suitability, grid infrastructure, and the scale of energy storage required. In the Hami region, as wind farm development expands, the sensitivity of wind energy output to wind speed fluctuations increases, a well-coordinated strategy to integrate wind farms with energy storage systems and grid infrastructure is crucial. Such integration could enhance grid flexibility, ensuring a stable energy supply despite fluctuations in wind energy generation. Policymakers must balance the initial construction costs and logistical challenges with the long-term benefits of higher wind speeds and energy generation potential.

This is especially important given that Hami plans to establish a significant computational power hub as part of the 'East Data West Computing' framework, which will consume a considerable amount of electricity (Wei 2024). The construction of large-scale intelligent computing centers is an energy-intensive operation, particularly in areas like Yiwu County in Hami, which has a favorable climate for cooling server rooms. This operation can be strategically aligned with the seasonal peak in wind power potential. Furthermore, the planned integration of Hami as a computational power auxiliary supply area, in conjunction with cities such as Zhengzhou, Henan and Qingyang, Gansu (Shang and Zhao 2024), emphasizes the significance of harnessing wind energy during its peak periods to support the demand of the growing computational power sector.

5. Conclusions

Our research provided a high-resolution assessment of wind energy potential in the Hami region of north-western China, based on integrating the ERA5 reanalysis dataset and advanced wind turbine technology. The findings of this research highlight several key aspects. First, the utilization of the high-resolution ERA5 reanalysis dataset addresses the limitations of previous studies that relied on lower-resolution data,

providing more accurate and localized assessments of wind energy potential. The comparison with observational data and previous studies demonstrates the importance of data resolution in accurately capturing wind speed patterns and potential energy generation. Second, the analysis at different hub heights reveals the significant impact of turbine height on wind energy capture. The power-law exponential distribution of wind profiles further emphasizes the need for tailored values to account for the complex wind behavior in the region. Finally, this study identifies the influence of wind turbine technology, including turbine scale and hub heights, on wind energy potential. The incorporation of specific turbine designs and layout considerations in the assessment enhances the precision of potential energy generation estimations.

Although our study demonstrates advancements in assessing wind energy potential, there are some limitations. For example, we focused on a specific region, which underscores the importance of conducting comprehensive assessments that integrate local data with broader-scale analyses. To create more accurate and comprehensive models of wind energy potential, future research can combine, future research should combine high-resolution data from specific areas like Hami with insights from large-scale assessments to create more accurate and comprehensive models of wind energy potential. Moreover, including terrain complexity, microclimatic variations, and land use patterns in wind resource assessments can improve the precision of renewable energy planning and development decisions.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-land?tab=overview>; <http://data.cma.cn/en/?r=data/>.

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